

History of Forensic Science

Forensic Science

According to the American Academy of
Forensic Sciences:

“The application of science to law”

Forensic Science

According to Richard Saferstein:

“Forensic science is the application of science to the criminal and
civil laws that are enforced by police agencies in a criminal
justice system”

According to Houck and Siegel:

“Forensic Science describes the science associating people,
places and things involved in criminal activities”

According to Houck (2011):

“Forensic Science is the science of spatial and temporal
relationships between people, place, and things involved in
crime.

Criminalistics = forensic science

Definitions are sometimes limiting as in this example of Forensic Science being applied in a non-criminal and non-civil event.



Latent Print Identification



- TEDAC device 482/04, a Long-Range Cordless Telephone, recovered near al-Qaim, Iraq on 4-17-2004.
- A latent print developed on this device matched a specific individual taken into custody for unrelated reasons on 4-22-2004.
- Investigation determined this individual was associated with a bomb making network and others identified.

TEDAC = Terrorist Explosives Device Analytical Center

Criminalistics – disciplines found in full-service forensic science laboratories that involve the examination, identification, and interpretation of physical evidence.

- Biological – serology and DNA
- Materials – trace evidence to include hairs, fibers, soil, building material
- Chemistry – drugs, explosives, tapes
- Pattern evidence (impression evidence) – documents, fingerprints, footwear, firearms
- Digital evidence – computers, cell phones

Forensic Science Specialties

- Forensic Pathology
- Forensic Entomology
- Forensic Odontology
- Forensic Anthropology
- Forensic Toxicology
- Forensic Engineering
- Microbial Forensics
- Forensic Art/Graphics
- WMD Forensics
- Wildlife Forensics
- Forensic Psychiatry/Psychology



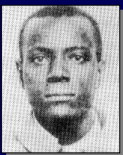
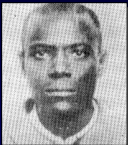
Innovations in Forensic Science

Alphonse Bertillon 1879 – personal identification through anthropometric measurements (22 measurements)



The West Case

Identification
Picture of
William West



Identification
Picture of
Will West

“West Brothers – Will and William West”

Early 1900's: A man was spotted in the incoming prisoner line at the U.S. Penitentiary in Leavenworth, Kansas by a guard Who recognized him and thought he was already in the prison population.

Upon examination, the incoming prisoner claimed to be named Will West, while the existing prisoner was named William West. According to their Bertillon measurements, they were essentially indistinguishable. Only their fingerprints could readily identify them, and the Bertillon Method was discredited. There is evidence that men named Will and William West were both imprisoned in the Federal Penitentiary in Leavenworth, Kansas, between 1903 and 1909.

Mathieu Orfila – father of Forensic Toxicology, French physician, and “medicolegalist”



1814 published first book on detection of poisons and their effects on animals

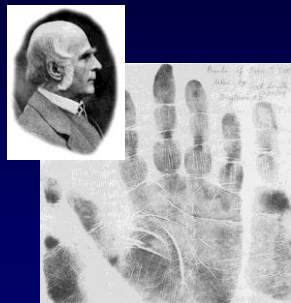
Innovations in Forensic Science

Francis Galton 1892 – identification through fingerprints

Permanence and Individualization

Formed near 12th week of gestation – completely formed by 18th week

Identical twins?

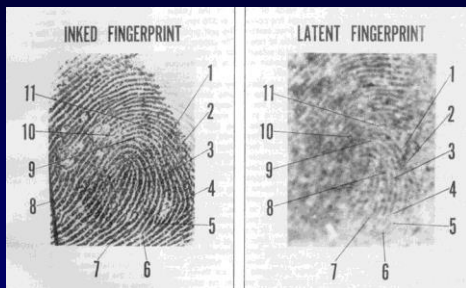


What is a “Latent” Fingerprint?

- Latent means hidden or invisible
- Sweat, oil, or other foreign substance
- Transfer to object when touched leaving recording of the ridges
- Visualized by powders, ALS and chemicals

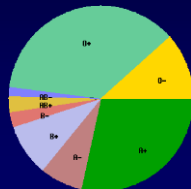
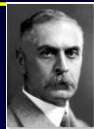


How are Identifications Made? “Galton” detail



Innovations in Forensic Science

- Karl Landsteiner 1909 – ABO system
- Leone Lattes 1915 – developed simple procedure to determine blood group (A, B, AB, O)



Serology Unit FBI Laboratory



FBI-5-178 A GENERAL VIEW OF THE SEROLOGY UNIT, PHYSICS AND CHEMISTRY SECTION, FBI LABORATORY, WASHINGTON, D. C.

FBI Laboratory 1932



FBI Laboratory Today

Innovations in Forensic Science

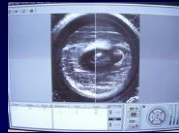
Calvin Goddard 1929 – comparison microscopy in ballistics
Started crime lab at Northwestern University, later became Chicago Police Department Lab (LAPD 1928, first Lab)





FBI-5-148 FBI LABORATORY EXAMINER PHOTOGRAPHS THE MARKINGS ON TEST AND EVIDENCE BULLETS

NIBIN – National Integrated Ballistics Information Network



Sniper Investigation October 2002

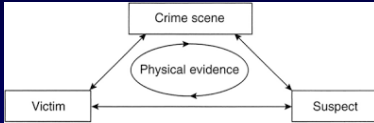




Innovations in Forensic Science

Edmond Locard – early 1900s

Locard's Exchange Principle: Whenever two objects come into contact with one another, there is exchange of Materials between them.



Lisk/Silva Investigation

1996 – 2002

REWARD
\$150,000.00
For the arrest and conviction
of the person or persons
responsible for the murders
of these children.

Kristin Lisk Kati Lisk Sofia Silva

1-800-729-1411

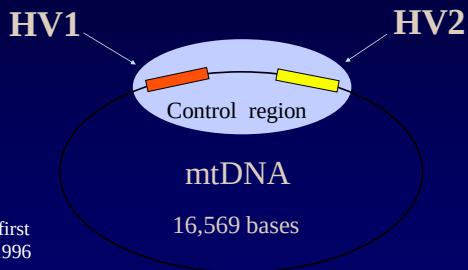
A SIX-YEAR SEARCH
A break in Lisk-Silva?

Rape suspect, 3 slayings in Virginia may be linked

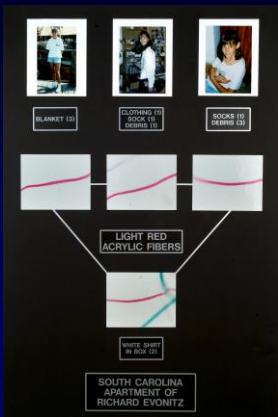
LAST DAYS OF RICHARD EVONITZ

The collage features several elements: a map of Virginia with locations like 'Lynchburg', 'Roanoke', and 'Charlottesville' marked; mugshots of individuals; and text from newspaper articles. One article mentions 'A SIX-YEAR SEARCH' and 'A break in Lisk-Silva?'. Another article is titled 'Rape suspect, 3 slayings in Virginia may be linked'. A third article is titled 'LAST DAYS OF RICHARD EVONITZ'. There are also photos of people and a scene that appears to be a crime scene or a location related to the investigation.

mtDNA match between Evonitz and a hair recovered from sock of Kati Lisk



mtDNA first used in 1996



Trace Evidence Links in the Silva/Lisk Investigation To Richard Evonitz

EVIDENCE	SOPHIA SILVA	KRISTIN LISK	KATI LISK
Head hair microscopically like Evonitz' known hair	X	X	X
Bedroom carpet fibers - Evonitz' Fredericksburg residence	X	X	X
Living room carpet fibers - Evonitz' Fredericksburg residence		X	
Computer room carpet fibers - Evonitz' Fredericksburg residence	X	X	X
Auto trunk fibers from Evonitz' 1992 Ford Taurus	X	X	X
Pink bath rug fibers - South Carolina search			X
Fibers from furry handcuffs - South Carolina	X	X	X
African fibers - South Carolina search	X		X
Dark Red acrylic fibers found in Evonitz' former residence and on numerous items from Evonitz	X	X	X
Light Red acrylic fibers found in pocket of white shirt - South Carolina search	X	X	X

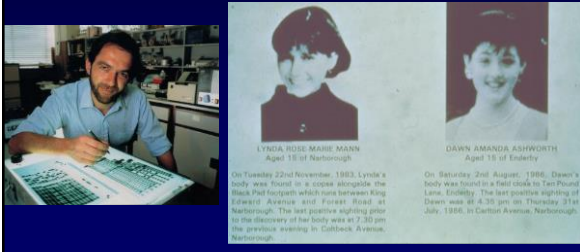
Latent fingerprint inside trunk after 6 years



Innovations in Forensic Science

Alec Jeffreys 1985 – DNA profiling (fingerprinting)

1983 and 1986 murders of two schoolgirls in England

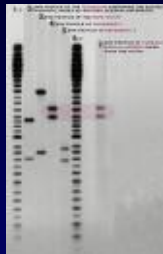


The Pitchfork Case

- Jeffreys was consulted in the case of the murder and rape of two British schoolgirls.
- Suspect cleared from analysis of semen samples at the scene and blood samples from the suspect.
- “Blooded” 4583 men
- Analysis of blood sample from Colin Pitchfork provided a match.

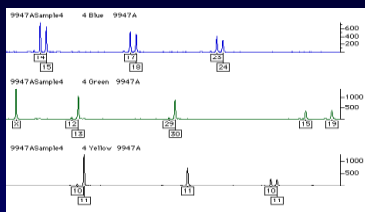
The Blooding by Joseph Wambaugh

DNA Profile 1988



FBI – December 1988

DNA Profile today



CODIS History

- DNA Identification Act of 1994 - formalized the FBI's authority to establish a national DNA Index System for law enforcement purposes.
- In October 1998 - the FBI's National DNA Index System (NDIS) became operational with 9 states participating in NDIS.
- CODIS statistics provided last week.

National DNA Database 1998



The Forensic Scientist is to provide truthful information in a judicial proceeding, not to prove guilt or innocence

- Frye vs US 1923 – court must decide whether the procedures or techniques are “generally accepted” by the relevant scientific community. (Polygraph)
- Federal Rules of Evidence (Rule 702) – scientific knowledge will assist the trier of fact to understand evidence if 1) the testimony is based upon sufficient data, 2) testimony is the product of reliable methods, and 3) witness has applied the methods reliably.

- Daubert v. Merrell Dow 1993 – Frye is not an absolute prerequisite to admissibility of scientific evidence. Gatekeeper determines
 - 1) has technique been tested (validation),
 - 2) subjected to peer review (published in literature)
 - 3) rate of error determined,
 - 4) standards exist, and
 - 5) has technique gained widespread acceptance in the relevant scientific community.
- Kumho Tire v. Carmichael 1999 – Gatekeeping role of judge applies not only to scientific testimony, but also to all expert testimony, i.e. scientific, technical and “other specialized knowledge.”
